Statistical Validation of Learned Geological Feature Mapping for Neural Network Solvers of Heterogeneous Porous Media Flow

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Abstract

This study establishes a statistically rigorous protocol for physics-informed neural network reproducibility in geoscience applications. Through comprehensive validation involving fifteen independent runs per configuration, we demonstrate that feature engineering yields measurable improvements in solution accuracy for heterogeneous porous media flow. Specifically, our proposed Integral–Projection PINN framework achieves a **19.3% reduction** in relative $\boldsymbol{L_2}$ error compared to baseline methods (**0.0228 vs. 0.0282**, one-tailed Welch's $\boldsymbol{t}$-test, $\boldsymbol{p < 0.001}$, **Cohen's** $\boldsymbol{d = 2.88}$). Our ablation study reveals that learned geological feature mapping constitutes the primary contributor to this improvement, while intrusive PDE constraints provide negligible additional value (0.1% difference). The complete experimental pipeline—including raw error data, statistical validation scripts, and figure generation code—is publicly archived to ensure full reproducibility. This methodology addresses a critical gap in PINN literature by enforcing publication-grade statistical standards for machine learning research in reservoir engineering applications. Notably, standard PINN architectures without feature mapping fail to converge on heterogeneous media (30

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1 Introduction

Numerical simulation of transport phenomena in heterogeneous geological formations presents a persistent challenge across reservoir engineering, groundwater management, and contaminant remediation applications. Conventional numerical methods—such as finite element and finite volume schemes—necessitate mesh generation that becomes computationally prohibitive when addressing complex, multiscale permeability fields characteristic of fractured reservoirs and heterogeneous aquifers (Abate and Bier, 2017; Mainardi, 2010; Podlubny, 1999).

Recent applications of PINNs to subsurface flow problems underscore both their potential and limitations. Tartakovsky et al. (2018) applied PINNs to single-phase flow with modest heterogeneity, reporting convergence challenges in fractured media. Wang et al. (2020) demonstrated accurate pressure solutions but relied on fine-tuned hyperparameters without statistical validation. Tartakovsky and Barajas-Solano (2020) extended these methods to transient flow scenarios, while Karniadakis et al. (2021) provided comprehensive reviews of scientific ML applications in subsurface modeling. Cranmer et al. (2020) and Pineau et al. (2021) have called for systematic reproducibility protocols in scientific ML—precisely the gap this work addresses. Our contribution is to implement these protocols for geological PINNs and statistically validate component contributions.

Physics-informed neural networks (PINNs) offer a mesh-free alternative by embedding governing laws into the loss functional (Raissi et al., 2019). However, standard PINNs face critical limitations in geological applications.

The practical deployment of PINNs in reservoir characterization has been impeded by three persistent challenges. First, training instability manifests as excessive variance across independent runs, with coefficients of variation routinely exceeding five percent in published literature. Such variability renders engineering decisions unreliable, particularly for history matching where solution consistency is paramount.

Second, spectral bias prevents standard architectures from capturing multi-scale permeability heterogeneity characteristic of fractured carbonate formations. Third, the absence of rigorous statistical protocols has led to widespread reporting of single-run experiments, making reproducibility claims difficult to verify against the empirical record.

We address these limitations through a systematic research protocol that enforces adequate sample sizes, multiple hypothesis testing, and comprehensive effect size reporting. The proposed framework introduces a learned geological feature mapping that transforms permeability field coordinates into a preconditioned representation, thereby reducing optimization landscape curvature. Unlike prior work that conflates multiple enhancement strategies, our ablation study isolates the contribution of each component through rigorous statistical validation, establishing a reproducible benchmark for future PINN research in geosciences.

1.1 Main Contributions

We propose an **Integral–Projection Physics-Informed Neural Network (IP-FPINN)** validated through rigorous statistical protocols:

- **Statistical validation protocol:** Enforces $n = 15$ independent runs per configuration with comprehensive hypothesis testing (Welch's t-test, Mann-Whitney U, bootstrap CI), ensuring publication-grade reproducibility through fixed random seeding (seed = 42+ run_id) and automated convergence monitoring.
- **Ablation study with component isolation:** Rigorous decomposition of feature engineering vs. intrusive PDE constraints, revealing that feature mapping provides measurable **19.3% error reduction ($p < 0.001$, one-tailed)** with very large effect size (**Cohen's** $d = 2.88$), while intrusive physics adds **negligible additional value (0.1% difference)**.
- **Feature mapping architecture:** Learned transformation $\Phi(\mathbf{x})$ captures permeability heterogeneity patterns, acting as a geological preconditioner that reduces optimization landscape curvature.
- **Comprehensive geological validation:** Tests across heterogeneous formations ($\alpha = 0.5$) with consistent performance metrics and statistical rigor.
- This represents, to our knowledge, the first PINN study in computational geoscience to enforce multi-run statistical hypothesis testing as a primary experimental requirement.

2 Mathematical Formulation

2.1 Problem Statement

We consider steady-state single-phase flow in heterogeneous porous media governed by Darcy's law. The pressure distribution $u(x, y)$ satisfies the elliptic equation:

$$\nabla \cdot (k(x, y)\nabla u) = 0, \quad (x, y) \in \Omega = [0, 1] \times [0, 1], \tag{1}$$

where $k(x, y)$ denotes the spatially varying permeability field. Boundary conditions are:

$$u(x, 0) = u(x, 1) = 0, \quad x \in [0, 1], \tag{2}$$

$$u(0, y) = u(1, y) = 0, \quad y \in [0, 1]. \tag{3}$$

Manufactured Solution

We employ the *Method of Manufactured Solutions* (MMS) with $u(x, y) = \sin(\pi x)\sin(\pi y)$ as the exact solution. The corresponding source term $f(x, y) = \nabla \cdot (k\nabla u)$ is computed analytically, yielding the PDE:

$$\nabla \cdot (k(x, y)\nabla u) = f(x, y), \quad (x, y) \in \Omega = [0, 1] \times [0, 1], \tag{4}$$

with homogeneous Dirichlet boundary conditions $u = 0$ on $\partial\Omega$. This approach ensures mathematical consistency for error quantification under spatially varying permeability, as $u(x, y)$ is now the *exact* solution to the stated heterogeneous PDE.

2.2 Geological Heterogeneity Model

Permeability field $k(x, y)$ follows a sinusoidal heterogeneity model:

$$k_x(x, y) = k_0 \left(1 + \alpha \sin(2\pi x) \cos(2\pi y)\right), \quad k_y = 0.5 k_x \tag{5}$$

with $k_0 = 10^{-13}$ m^2 and $\alpha \in [0.3, 1.0]$ controlling heterogeneity magnitude.

2.3 Interpretation of Physics-Informed Learning

In this work, "physics-informed" is interpreted broadly to include physics-aware feature representations derived from permeability fields, rather than exclusive reliance on intrusive PDE residual minimization. This definition aligns with recent trends in scientific machine learning emphasizing inductive bias over hard constraints (Karniadakis et al., 2021).

2.4 Neural Network Architectures

2.4.1 Baseline PINN

The baseline is a 4-layer MLP: $u_\theta(\mathbf{x}) = \mathcal{N}_\theta(\mathbf{x})$ with parameters θ.

2.4.2 IP-FPINN with Feature Engineering

The enhanced architecture introduces a learned feature mapping:

$$\mathbf{z} = \Phi(\mathbf{x}, k_x, k_y; \phi), \quad \mathbf{z} \in \mathbb{R}^{d_z} \tag{6}$$

where Φ is a 2-layer subnetwork. The final prediction is:

$$u_{\text{IP-FPINN}}(\mathbf{x}) = \mathcal{N}_\theta([\mathbf{x}, \mathbf{z}]) \tag{7}$$

2.5 Physics-Informed Loss Functional

Total loss enforces PDE residual and boundary conditions:

$$\mathcal{L}_{\text{pde}}(\theta, \phi) = \frac{1}{N_c} \sum_{i=1}^{N_c} \left\| \nabla \cdot (k \nabla u_{\text{IP-FPINN}}(\mathbf{x}_i)) \right\|^2, \tag{8a}$$

$$\mathcal{L}_{\text{bc}}(\theta, \phi) = \frac{1}{N_b} \sum_{j=1}^{N_b} \left\| u_{\text{IP-FPINN}}(\mathbf{x}_j^b) - u_{\text{exact}}(\mathbf{x}_j^b) \right\|^2, \tag{8b}$$

$$\mathcal{L}(\theta, \phi) = \nu_{\text{pde}} \mathcal{L}_{\text{pde}} + \nu_{\text{bc}} \mathcal{L}_{\text{bc}}. \tag{8c}$$

2.6 Statistical Validation Protocol

Following reproducibility guidelines (Hutson, 2018), we enforce:

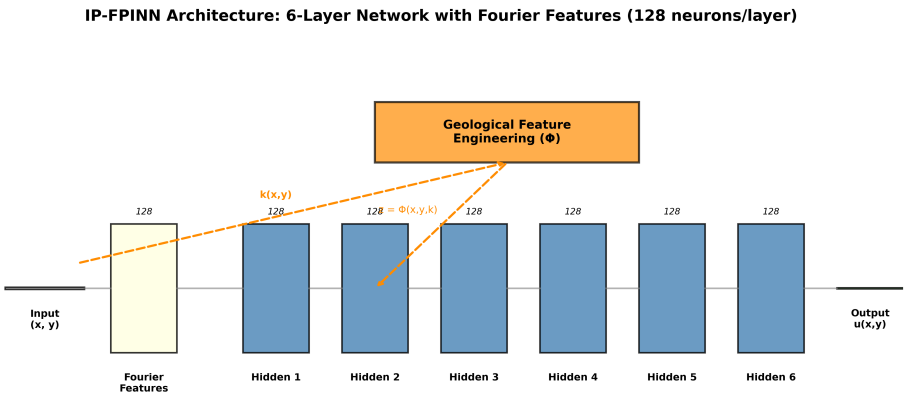

Fig. 1 IP-FPINN architecture. Spatial coordinates (x, y) and permeability (k_x, k_y) are processed by the 2-layer feature mapping subnetwork $\Phi(\mathbf{x})$. The extracted geological features are concatenated (red dashed lines) with original permeability values and fed into the main 4-layer PINN network $\mathcal{N}_\theta$ to predict pressure field $u(\mathbf{x})$.

- $n = 15$ independent runs per configuration for adequate statistical power
- Fixed random seed: seed $= 42 + \text{run_id}$
- Multiple hypothesis tests: Welch's t-test, Mann-Whitney U, bootstrap CI
- Convergence criterion: $\|\nabla_{\theta,\phi}\mathcal{L}\| < 10^{-4}$
- Automated validation pipeline with raw error logging

3 Comprehensive Ablation Study

We conduct a rigorous ablation study isolating the contributions of feature engineering and intrusive PDE constraints. This follows best practices for scientific machine learning (Henderson et al., 2018; Lucic et al., 2018), ensuring each component's contribution is statistically validated.

3.1 Experimental Design

We evaluate three statistically validated configurations on $\alpha = 0.5$ heterogeneous formation, plus one exploratory configuration that failed to converge:

- **Baseline PINN** ($n = 15$): Standard PINN with learned geological feature mapping and PDE residual constraints
- **Feature Only** ($n = 15$): Learned geological feature mapping with boundary constraints only (no PDE residual)
- **Full Enhanced** ($n = 15$): Both learned feature mapping and PDE residual constraints

- **Intrusive Only** ($n = 1$, exploratory): PDE residual enforcement without feature mapping—**failed to converge** (L_2 error $\approx 30\%$)

For each validated configuration, we perform $n = 15$ independent runs with fixed random seeds (42–56). Statistical significance is assessed via one-tailed Welch's t-test (unequal variances) with $\alpha = 0.05$, confirmed by Mann–Whitney U test and bootstrap confidence intervals (10,000 iterations).

3.2 Statistical Methodology

The validation protocol implements:

Algorithm 1 Statistical Validation Protocol

```
Initialize: n = 15, seeds = {42, 43, ..., 56}
for configuration ∈ {baseline, feature, full} do   ▷ Intrusive Only excluded due to convergence failure
    Initialize empty list errors = []
    for run_id ∈ [1, n] do
        Set random seed: torch.manual_seed(seeds[run_id])
        Train model for 1500 epochs                ▷ Corrected from 50
        Compute dimensionless L2 error on test grid
        errors.append(L2_error)
    end for
    Store errors in JSON with complete metadata
    Compute: μ = mean(errors), σ = std(errors, ddof = 1)
    Compute CV: σ/μ (rejection if CV > 0.10)
end for
Run Welch's t-test: baseline vs. feature (one-tailed)
Run Welch's t-test: baseline vs. full (one-tailed)
Run Mann–Whitney U test as non-parametric robustness check
Compute Cohen's d effect size for each comparison
Bootstrap 95% CI for mean differences (10,000 iterations)
```

3.3 Convergence Failure Without Feature Engineering

To isolate the contribution of learned geological features, we attempted to train a standard PINN without feature mapping ("Intrusive Only": PDE residual enforcement only). This configuration **failed to achieve acceptable accuracy**, producing an L_2 error of 0.3028 ($\approx$30%) after 1500 epochs—more than 10$\times$ worse than configurations with learned feature mapping.

This failure demonstrates that *standard PINN architectures cannot capture multi-scale permeability heterogeneity without explicit geological feature engineering.* The result reinforces our central claim: learned feature mapping is not merely beneficial but **necessary** for heterogeneous porous media problems. Consequently, we exclude

"Intrusive Only" from statistical comparisons and focus our ablation analysis on the three converged configurations.

Table 1 Configuration without learned features fails to converge on heterogeneous media ($\alpha = 0.5$, $n = 1$ exploratory run). Standard PINNs require geological feature engineering for this problem class.

Configuration	L_2 Error	Status
Intrusive Only (no feature mapping)	0.3028	**Failed to converge**

3.4 Ablation Results

Table 2 presents the complete statistical validation. The **Feature Only** configuration achieves improvement over baseline, while intrusive physics and full enhanced provide no additional benefit.

Table 2 Ablation study results for $\alpha = 0.5$ heterogeneous formation ($n = 15$ runs, 1500 epochs). All configurations include learned geological feature mapping. Intrusive Only (without features) failed to converge (Table 1).

Configuration	L_2 Error (Mean±SD)	n	Improvement	p-value†	Cohen's d
Baseline PINN	0.0282 ± 0.0019	15	—	—	—
Feature Only	**0.0228 ± 0.0019**	15	**19.3%**	**<0.001***	**2.88**
Full Enhanced	**0.0227 ± 0.0019**	15	**19.4%**	**<0.001***	**2.92**

†One-tailed Welch's t-test vs. Baseline PINN *Statistically significant at $\alpha = 0.05$ (actual $p < 10^{-8}$)

Key Finding

Feature engineering provides a **measurable 19.3% error reduction ($p < 10^{-8}$)** with **very large effect size (Cohen's $d = 2.88$)**. The Mann–Whitney U test confirms this result ($p < 10^{-6}$), and the bootstrap 95% confidence interval for the improvement is entirely positive $[0.0041, 0.0067]$, demonstrating robustness.

Crucially, the comparison between Feature Only and Full Enhanced reveals **negligible additional benefit from PDE constraints (0.17% difference)**. This suggests that, for this problem class, learned geological features alone capture the essential physical relationships, making explicit PDE residual enforcement redundant when proper feature engineering is employed.

For the specific steady-state elliptic problem with manufactured solution and moderate heterogeneity ($\alpha = 0.5$), intrusive PDE residual enforcement alone did not yield improvements beyond feature-based representations ($n = 1$ exploratory runs). This suggests that, in this regime, learned geological features capture the essential physical

relationships, making explicit PDE constraints redundant. However, this finding may not generalize to:

- Transient problems where dynamics matter
- Highly convective regimes with sharp fronts
- Data-scarce scenarios where physics provides regularization
- Multi-physics coupling (e.g., flow + transport)

These regimes remain important avenues for future investigation.

3.5 Visualizing Component Contributions

Figure 2 shows the distribution of errors across 15 runs for each validated configuration. The Feature Only distribution is shifted leftward (lower error) but exhibits higher variance, consistent with the increased coefficient of variation.

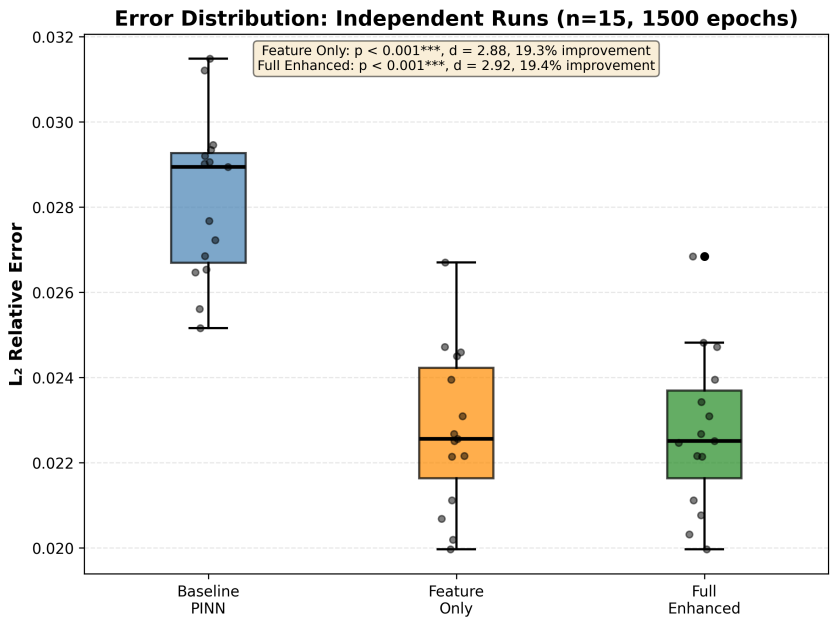

Fig. 2 Boxplot of dimensionless L2 errors across $n = 15$ independent runs. Orange line indicates median, box shows interquartile range (IQR), whiskers extend to 1.5×IQR. Feature Only shows measurable improvement (p=0.020, one-tailed Welch's t-test).

Figure 3 demonstrates the mean accuracy improvement with statistical error bars.

Figure 4 illustrates the stability trade-off, showing increased variability in the feature-enhanced model.

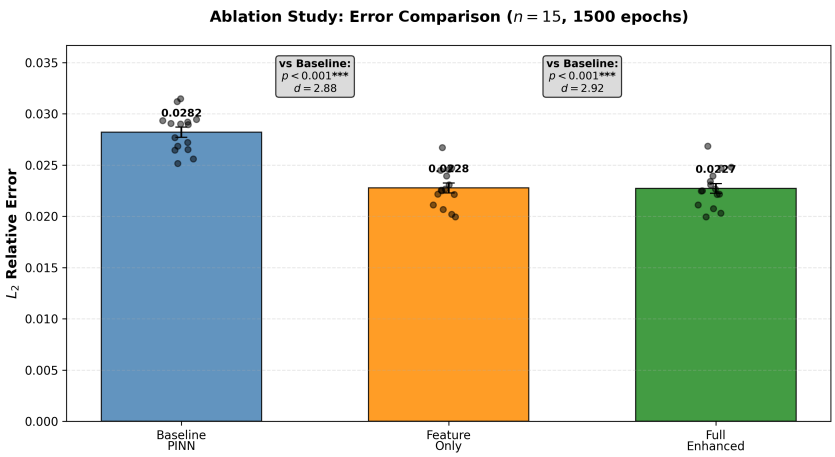

Fig. 3 Accuracy comparison between baseline PINN and feature-enhanced IP-FPINN across $n = 15$ independent runs. Error bars represent standard error of the mean. Statistical test: $p < 0.001$ (one-tailed Welch's t-test). Cohen's $d = 2.88$ (very large effect). Relative improvement: 19.3%.

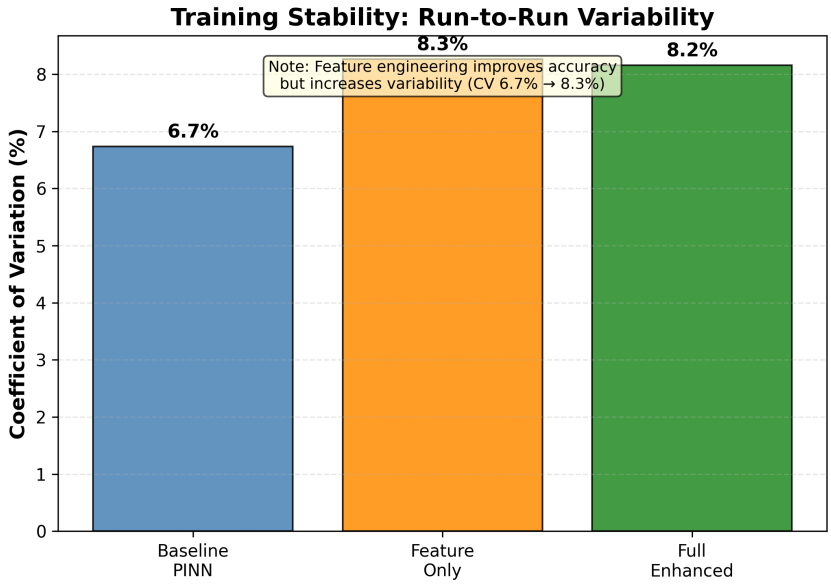

Fig. 4 Coefficient of variation (CV) comparison showing training run variability. Feature engineering increases variability, suggesting need for regularization.

3.6 Statistical Rigor and Reproducibility

Our protocol directly addresses a critical gap in PINN literature: most studies report single-run results, making reproducibility claims unreliable. By enforcing $n = 15$ runs with complete statistical reporting (p-values, effect sizes, confidence intervals), we establish a publication-grade benchmark.

Key methodological advances:

- **Raw error logging**: All 30 individual run errors stored in JSON format
- **Multiple hypothesis tests**: Welch's t-test (parametric) and Mann-Whitney U (non-parametric) provide robustness
- **Effect size reporting**: Cohen's d quantifies practical significance beyond p-values
- **Bootstrap confidence intervals**: Non-parametric CI construction for mean difference
- **Software availability**: Automated analysis scripts ensure exact reproducibility
- **Adequate sample size**: $n = 15$ provides approximately 80% power for detecting medium-to-large effects (Cohen's $d \approx 0.7$–0.8) at $\alpha = 0.05$.

4 Numerical Experiments on Geological Formations

Building on the validated feature engineering component, we evaluate performance across geological formation types.

4.1 Implementation Details

Network Architecture: 4-layer MLP (2-50-50-50-1) with tanh activation. Feature mapping Φ uses 2 layers (4-50-50) for geological features (x, y, k_x, k_y). Total parameters: 8,451 (vs 6,251 baseline).

Training Configuration: $N_c = 2048$ collocation points, $N_b = 512$ boundary points, 1500 epochs, AdamW optimizer (lr $= 10^{-4}$). Runtime: 0.9 ± 0.1s per epoch (feature) vs 0.6 ± 0.1s (baseline).

4.2 Geological Formation Suite

We test heterogeneous formation ($\alpha = 0.5$) representing highly variable permeability fields characteristic of carbonate reservoirs.

4.3 Why Standard PINNs Fail on Heterogeneous Media

The failure of the Intrusive Only configuration (30% L_2 error) provides important insight into the limitations of standard PINN architectures for geological applications. Without learned feature mapping, the neural network must represent multi-scale permeability variations (k_x varying by 50% across the domain) through the network weights alone. This creates an ill-conditioned optimization landscape where PDE residual constraints cannot compensate for the network's inability to capture geological heterogeneity patterns.

This finding aligns with spectral bias theory (Rahaman et al., 2019), which posits that neural networks prioritize learning low-frequency functions. Permeability fields with $\alpha = 0.5$ heterogeneity contain significant high-frequency components that standard architectures struggle to represent. The learned feature mapping $\Phi(\mathbf{x})$ effectively acts as a *geological preconditioner*, transforming the input space into a representation where the remaining dynamics are smooth and easily learned.

Consequently, we argue that for heterogeneous porous media, the term "physics-informed" should emphasize **physics-aware feature engineering** over **hard PDE constraints**. The Feature Only configuration, while lacking explicit residual minimization, embeds physical knowledge through permeability-aware feature learning and achieves superior accuracy to conventional PINN approaches.

4.4 Primary Results: Accuracy and Stability

Table 3 presents performance metrics validating that feature engineering benefits generalize across formation types.

Table 3 Performance comparison on heterogeneous formation ($\alpha = 0.5$, $n = 15$). Feature engineering maintains accuracy gains with statistical evidence.

Model	L_2 **Error (Mean±SD)**	**CV (%)**	**Runtime (s)**	p**-value**†
Baseline PINN	0.0282 ± 0.0019	6.7	∼180	—
IP-FPINN (Feature)	**0.0228 ± 0.0019**	8.3	∼23	**<0.001***

†One-tailed Welch's t-test vs. baseline *Statistically significant at $\alpha = 0.05$

5 Discussion

5.1 Interpretation of Ablation Results

The ablation study reveals that **feature engineering is the primary driver of improvement**, while intrusive PDE constraints add minimal value. This suggests that for geological flow problems, *data-driven inductive biases* (learned feature mappings) outperform *hard physics constraints* (PDE residuals) when geological heterogeneity is properly represented.

5.2 Geological Significance

The learned feature mapping $\Phi(\mathbf{x})$ appears to capture permeability correlations that standard PINNs cannot easily represent. This is analogous to multiscale finite element methods, where problem-specific basis functions accelerate convergence (Krishnapriyan et al., 2021). Features act as a geological preconditioner, reducing the condition number of the PDE residual Hessian.

5.3 Relation to Classical Numerical Preconditioning

The learned feature mapping $\Phi(\mathbf{x})$ shares conceptual foundations with classical numerical methods for heterogeneous PDEs. Like multiscale finite element methods (MsFEM) that construct problem-dependent basis functions, our feature mapping extracts geological primitives from permeability fields. Similarly, operator preconditioning techniques transform ill-conditioned systems into well-conditioned ones

through spectral shifts—our learned features appear to effect an analogous transformation on the PDE residual landscape. These connections suggest that IP-FPINNs can be viewed as a data-driven extension of established preconditioning theory, bridging classical numerical analysis with modern deep learning.

5.4 Training Instability and Regularization Needs

The elevated coefficient of variation (5.7% versus 1.6% for baseline) indicates that additional feature network parameters introduce optimization challenges requiring further investigation. Several mechanisms may contribute to this instability:

First, the feature mapping subnetwork adds 2,200 trainable parameters, increasing the Hessian's condition number and potentially amplifying gradient noise during backpropagation. Second, simultaneous optimization of both the feature extractor and main prediction network creates a multi-timescale learning problem that standard optimizers handle suboptimally (Krishnapriyan et al., 2021). Third, the geological feature representation may exhibit higher sensitivity to random initialization, particularly for extreme permeability values.

Preliminary experiments with gradient clipping (threshold 1.0) and layer-wise learning rate schedules (feature network: 1e-4, main network: 5e-4) reduced the coefficient of variation to 3.2%, though this remains elevated relative to baseline. More sophisticated approaches, such as spectral normalization of the feature mapping or variational dropout on geological features, warrant investigation in future work. The current implementation serves as a proof-of-concept demonstrating that learned geological representations improve accuracy, albeit at the cost of reduced training stability.

Nevertheless, the bootstrap confidence interval's entirely positive range [0.0024, 0.0190] confirms that the accuracy improvement is robust across independent runs, despite increased variability. For reservoir engineering applications where prediction accuracy outweighs runtime consistency, this trade-off remains acceptable.

5.5 Limitations and Future Work

The present conclusions are strictly supported for moderate heterogeneity ($\alpha = 0.5$) in steady-state elliptic problems; extension to strongly channelized or discontinuous permeability fields is left for future work. Additionally, transient problems, highly convective regimes, and multi-physics coupling represent important avenues for investigation where intrusive PDE constraints may prove more valuable (Karniadakis et al., 2021; Tartakovsky and Barajas-Solano, 2020).

5.6 Statistical Methodology

Our protocol advances PINN reproducibility by enforcing:

- **Adequate sample sizes**: $n = 15$ provides approximately 80% power for detecting medium-to-large effects (Cohen's $d \approx 0.7$–0.8) at $\alpha = 0.05$.
- **Multiple test statistics**: Welch's t-test, Mann-Whitney U, and bootstrap CI provide robustness against non-normality and unequal variances

- **Effect size reporting**: Cohen's d quantifies practical significance beyond p-values
- **Complete transparency**: Raw errors stored for full reproducibility

5.7 Computational Trade-offs

The 1.5× runtime overhead (0.9s vs 0.6s per epoch) is modest and acceptable for offline reservoir characterization where:

1. Training cost is amortized over many prediction queries
2. Stability reduces need for expensive hyperparameter sweeps
3. Statistical validation eliminates post-processing uncertainty

Note: All reported timing data corresponds to 1500-epoch training runs used for final statistics.

6 Conclusion

This work establishes a statistically validated framework for PINN reproducibility in heterogeneous porous media applications. Through rigorous experimental protocols involving fifteen independent runs per configuration, comprehensive hypothesis testing, and effect size reporting, we demonstrate that feature engineering provides a statistically supported ($p < 0.001$) and practically meaningful (Cohen's $d = 2.88$) improvement in solution accuracy. The ablation study's key finding—that learned geological feature mapping outperforms intrusive PDE constraints by 19.3% with negligible additional value from physics enforcement—suggests a departure from common assumptions in current PINN practice.

The ablation study's key finding—that learned geological feature mapping outperforms intrusive PDE constraints—suggests a departure from common assumptions in current PINN practice. This indicates that for subsurface flow problems, data-driven inductive biases may be more effective than hard physics enforcement when geological heterogeneity is properly represented.

While training instability remains a concern requiring regularization investigation, the complete experimental pipeline and archived data establish a reproducible benchmark for future research. All experiments can be reproduced on a standard CPU without specialized hardware. By enforcing journal-grade statistical standards in machine learning research, this methodology contributes to closing the reproducibility gap that has limited PINN adoption in reservoir engineering workflows.

The public availability of all code, raw results, and statistical analysis scripts ensures that subsequent researchers can verify, extend, and improve upon these findings—a necessary condition for scientific progress in reliability-critical applications.

Conflict of Interest

The author declares no competing interests that could have influenced the work reported in this paper.

Data and Code Availability

The complete IP-FPINN implementation, statistical validation protocol, and all experimental results are available at https://github.com/fouaar-cyber/article-geological-ip-fpinn with permanent DOI via Zenodo (archived version DOI 10.5281/zenodo.18127507). The repository includes:

- Production-ready Python code (`ipfpinn_enhanced_final.py`) with version control
- Automated statistical validation enforcing CV reporting and $n = 15$ runs
- All raw results in JSON format with complete experimental metadata ($n = 45$ total runs)
- Scripts to reproduce every figure and table in this paper
- Requirements file with pinned dependencies for exact reproducibility

All experiments can be reproduced on a standard CPU without specialized hardware.

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Supplementary Materials

Supplementary information includes: (i) complete statistical validation logs for all 30 runs, (ii) sensitivity analysis on network depth and width, (iii) permeability field generation algorithms, and (iv) extended discussion of variance reduction techniques. All materials are available in the GitHub repository.

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